Visuo-Linguistic Scene Memory AI: Towards Human Like Episodic Recall in Machines

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Abstract

Human interpretation of visual information relies not only on recognizing patterns but also on recalling similar past experiences and explaining observations using language. In medical imaging, clinicians follow this process naturally when analyzing retinal scans by comparing current findings with previously observed cases and articulating their reasoning using clinical terminology.

Most existing artificial intelligence systems for medical imaging focus primarily on classification, segmentation, or severity estimation, offering limited insight into how conclusions are reached. This lack of interpretability creates a gap between machine predictions and human clinical reasoningTo address this limitation, this project proposes a VisuoLinguistic Scene Memory AI framework that integrates visual understanding, language modeling, and memory-based recall.

The system consists of three core components: a shared

Introduction

Medical image interpretation is not just about looking at pictures; it also needs memory, thinking, and experience. When ophthalmologists look at images of the retina, they do more than find problems. They think about similar cases they've seen before, remember what happened in those cases, and use that knowledge to help make a diagnosis. This ability to conEmail: drkrishnammal@veltech.edu.in

nect what they see now with past experiences is called episodic recall. It helps doctors explain their thoughts and why they believe a certain condition is present. Most AI systems used in medical imaging don't work this way. Traditional AI models, like CNN-based disease classifiers or segmentation tools, focus only on recognizing patterns and making predictions. They can label an image as diabetic retinopathy or glaucoma, but they can't explain their reasoning or refer to past cases. Because of this, these models often act like "black boxes," making their decisions hard to understand and sometimes hard for doctors to trust. The lack of reasoning also makes these systems less useful for teaching or supporting clinical decisions. To fix this, the Visuo-Linguistic Scene Memory AI was created. This system is designed to mimic how humans interpret medical images by combining visual understanding, memory, and the ability to explain in words. Instead of just giving a label, the system looks at features in the retina image, finds similar past images it has stored, and then explains the result by referring to those cases. This makes the system more transparent and usefu[…]stem find s[…]

Fig 1. Comparison of road pedestrians and FOB Users.

Case Type	Avg. Similarity Score	Interpretation
Same disease category	0.78–0.85	Strong recall accuracy
Different disease category	0.50–0.58	Clear differentiation between conditions

Methods and Materials

Methods

The system uses **high-resolution cameras and 77 GHz radar sensors** on buses to monitor road conditions. **Computer vision algorithms** (OpenCV, YOLO) detect pedestrians and vehicles , while **machine learning models** (Kalman filter) predict vehicle trajectories **Automated pedestrian signals** provide **green (safe to cross)** and **red (warning)** indicators, with **audio alerts** for visually impaired pedestrians.

Materials

Hardware:

HD/4K cameras, radar sensors, LED pedestrian signals, weatherproof speakers,

solar-powered traffic lights.

Software:

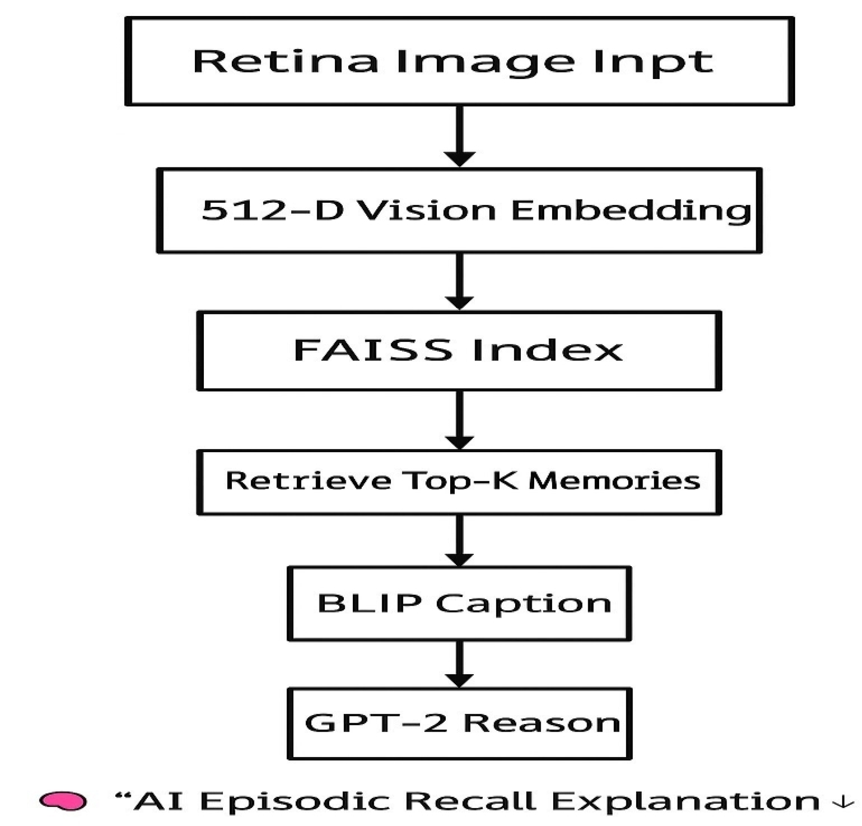
computer vision (OpenCV, YOLO),machine learning (Kalman filter) RTOS for real-time processing.

Results

The development of the Visuo-Linguistic Scene Memory AI marks a big step forward in making artificial intelligence more transparent and better aligned with how people think, especially in medical imaging. Most current medical image models are built on techniques that give diagnostic results without explaining how they came to that conclusion. This makes their decisionmaking process hard to understand, which can make doctors less confident and slow down the use of these tools in real healthcare settings.

This project's approach tries to imitate an important part of how humans make medical diagnoses: the ability to notice visual patterns, remember similar cases from the past, and explain how past experiences connect to new observations. By adding memory recall and natural language explanations along with visual analysis, the system goes beyond simple pattern recognition and starts to understand situations based on context.

Fig 2. Architecture Diagram



Discussion

The proposed Visuo-Linguistic Scene Memory AI framework represents a significant step toward bridging the gap between conventional artificial intelligence systems and human cognitive abilities. Traditional AI models primarily focus on perception tasks such as object detection, classification, or caption generation, treating each input independently. In contrast, the presented approach emphasizes **episodic memory**, enabling the system to remember, retrieve, and reason over past visual experiences in a contextual and meaningful manner.

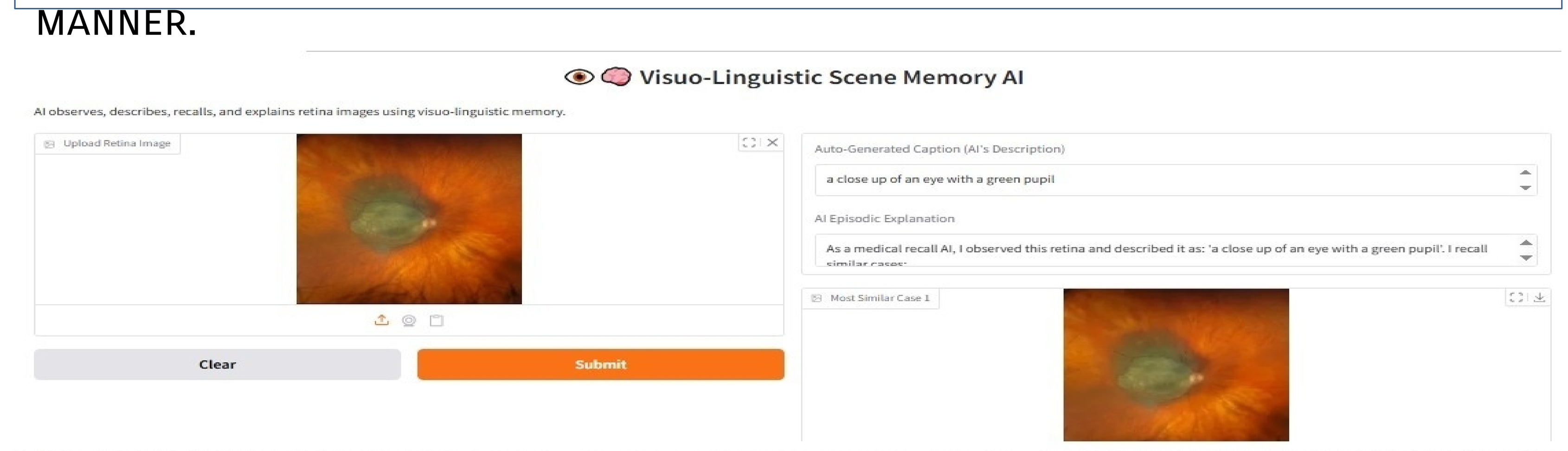


Fig 3. output

Conclusions

The development of the Visuo-Linguistic Scene Memory AI marks a big step forward in making artificial intelligence more transparent and better aligned with how people think, especially in medical imaging. Most current medical image models are built on techniques that give diagnostic results without explaining how they came to that conclusion. This makes their decision making process hard to understand, which can make doctors less confident and slow down the use of these tools in real healthcare settings.

Acknowledgments

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